3.0 INTRODUCTION

3.3 Limitations of Reliability Predictions - This handbook provides a common basis for reliability predictions, based on analysis of the best available data at the time of issue. It is intended to make reliability prediction as good a tool as possible. However, like any tool, reliability prediction must be used intelligently, with due consideration of its limitations.

The first limitation is that the failure rate models are point estimates which are based on available data. Hence, they are valid for the conditions under which the data was obtained, and for the devices covered. Some extrapolation during model development is possible, but the inherently empirical nature of the models can be severely restrictive. For example, none of the models in this handbook predict nuclear survivability or the effects of ionizing radiation.

Even when used in similar environments, the differences between system applications can be significant. Predicted and achieved reliability have always been closer for ground electronic systems than for avionic systems, because the environmental stresses vary less from system to system on the ground and hence the field conditions are in general closer to the environment under which the data was collected for the prediction model. However, failure rates are also impacted by operational scenarios, operator characteristics, maintenance practices, measurement techniques and differences in definition of failure. Hence, a reliability prediction should never be assumed to represent the expected field reliability as measured by the user (i.e., Mean-Time-Between-Maintenance, Mean-Time-Between-Removals, etc.). This does not negate its value as a reliability engineering tool; note that none of the applications discussed above requires the predicted reliability to match the field measurement.

Electronic technology is noted for its dynamic nature. New types of devices and new processes are continually introduced, compounding the difficulties of predicting reliability. Evolutionary changes may be handled by extrapolation from the existing models; revolutionary changes may defy analysis.

Another limitation of reliability predictions is the mechanics of the process. The part stress analysis method requires a significant amount of design detail. This naturally imposes a time and cost penalty. More significantly, many of the details are not available in the early design stages. For this reason this handbook contains both the part stress analysis method (Sections 5 through 23) and a simpler parts count method (Appendix A) which can be used in early design and bid formulation stages.

Finally, a basic limitation of reliability prediction is its dependence on correct application by the user. Those who correctly apply the models and use the information in a conscientious reliability program will find the prediction a useful tool. Those who view the prediction only as a number which must exceed a specified value can usually find a way to achieve their goal without any impact on the system.

3.4 Part Stress Analysis Prediction

3.4.1 Applicability - This method is applicable when most of the design is completed and a detailed parts list including part stresses is available. It can also be used during later design phases for reliability trade-offs vs. part selection and stresses. Sections 5 through 23 contain failure rate models for a broad variety of parts used in electronic equipment. The parts are grouped by major categories and, where appropriate, are subgrouped within categories. For mechanical and electromechanical parts not covered by this Handbook, refer to Bibliography items 20 and 36 (Appendix C).

The failure rates presented apply to equipment under normal operating conditions, i.e., with power on and performing its intended functions in its intended environment. Extrapolation of any of the base failure rate models beyond the tabulated values such as high or sub-zero temperature, electrical stress values above 1.0, or extrapolation of any associated model modifiers is completely invalid. Base failure rates can be interpolated between electrical stress values from 0 to 1 using the underlying equations.

The general procedure for determining a board level (or system level) failure rate is to sum individually calculated failure rates for each component. This summation is then added to a failure rate for the circuit board (which includes the effects of soldering parts to it) using Section 16, Interconnection Assemblies.

For parts or wires soldered together (e.g., a jumper wire between two parts), the connections model appearing in Section 17 is used. Finally, the effects of connecting circuit boards together is accounted for by adding in a failure rate for each connector (Section 15, Connectors). The wire between connectors is assumed to have a zero failure rate. For various service use profiles, duty cycles and redundancies the procedures described in MIL-STD-756, Reliability Modeling and Prediction, should be used to determine an effective system level failure rate.

3.4.2 Part Quality - The quality of a part has a direct effect on the part failure rate and appears in the part models as a factor, π_Q. Many parts are covered by specifications that have several quality levels, hence, the part models have values of π_Q that are keyed to these quality levels. Such parts with their quality designators are shown in Table 3-1. The detailed requirements for these levels are clearly defined in the applicable specification, except for microcircuits. Microcircuits have quality levels which are dependent on the number of MIL-STD-883 screens (or equivalent) to which they are subjected.

Table 3-1: Parts With Multi-Level Quality Specifications

Part	Quality Designators
Microcircuits	S, B, B-1, Other: Quality Judged by Screening Level
Discrete Semiconductors	JANTXV, JANTX, JAN
Capacitors, Established Reliability (ER)	D, C, S, R, B, P, M, L
Resistors, Established Reliability (ER)	S, R, P, M
Coils, Molded, R.F., Reliability (ER)	S, R, P, M
Relays, Established Reliability (ER)	R, P, M, L

Some parts are covered by older specifications, usually referred to as Nonestablished Reliability (Non-ER), that do not have multi-levels of quality. These part models generally have two quality levels designated as "MIL-SPEC.", and "Lower". If the part is procured in complete accordance with the applicable specification, the π_Q value for MIL-SPEC should be used. If any requirements are waived, or if a commercial part is procured, the π_Q value for Lower should be used.

The foregoing discussion involves the "as procured" part quality. Poor equipment design, production, and testing facilities can degrade part quality. The use of the higher quality parts requires a total equipment design and quality control process commensurate with the high part quality. It would make little sense to procure high quality parts only to have the equipment production procedures damage the parts or introduce latent defects. Total equipment program descriptions as they might vary with different part quality mixes is beyond the scope of this Handbook. Reliability management and quality control procedures are described in other DoD standards and publications. Nevertheless, when a proposed equipment development is pushing the state-of-the-art and has a high reliability requirement necessitating high quality parts, the total equipment program should be given careful scrutiny and not just

the parts quality. Otherwise, the low failure rates as predicted by the models for high quality parts will not be realized.

3.4.3 Use Environment - All part reliability models include the effects of environmental stresses through the environmental factor, π_E, except for the effects of ionizing radiation. The descriptions of these environments are shown in Table 3-2. The π_E factor is quantified within each part failure rate model. These environments encompass the major areas of equipment use. Some equipment will experience more than one environment during its normal use, e.g., equipment in spacecraft. In such a case, the reliability analysis should be segmented, namely, missile launch (M_L) conditions during boost into and return from orbit, and space flight (S_F) while in orbit.

Table 3-2: Environmental Symbol and Description

Environment	π_E Symbol	Equivalent MIL-HDBK-217E, Notice 1 π_E Symbol	Description
Ground, Benign	G_B	G_B G_{MS}	Nonmobile, temperature and humidity controlled environments readily accessible to maintenance; includes laboratory instruments and test equipment, medical electronic equipment, business and scientific computer complexes, and missiles and support equipment in ground silos.
Ground, Fixed	G_F	G_F	Moderately controlled environments such as installation in permanent racks with adequate cooling air and possible installation in unheated buildings; includes permanent installation of air traffic control radar and communications facilities.
Ground, Mobile	G_M	G_M M_P	Equipment installed on wheeled or tracked vehicles and equipment manually transported; includes tactical missile ground support equipment, mobile communication equipment, tactical fire direction systems, handheld communications equipment, laser designations and range finders.
Naval, Sheltered	N_S	N_S N_{SB}	Includes sheltered or below deck conditions on surface ships and equipment installed in submarines.
Naval, Unsheltered	N_U	N_U N_{UU} N_H	Unprotected surface shipborne equipment exposed to weather conditions and equipment immersed in salt water. Includes sonar equipment and equipment installed on hydrofoil vessels.

Table 3-2: Environmental Symbol and Description (cont'd)

Environment	π_E Symbol	Equivalent MIL-HDBK-217E, Notice 1 π_E Symbol	Description
Airborne, Inhabited, Cargo	A_{IC}	A_{IC} A_{IT} A_{IB}	Typical conditions in cargo compartments which can be occupied by an aircrew. Environment extremes of pressure, temperature, shock and vibration are minimal. Examples include long mission aircraft such as the C130, C5, B52, and C141. This category also applies to inhabited areas in lower performance smaller aircraft such as the T38.
Airborne, Inhabited, Fighter	A_{IF}	A_{IF} A_{IA}	Same as A_{IC} but installed on high performance aircraft such as fighters and interceptors. Examples include the F15, F16, F111, F/A 18 and A10 aircraft.
Airborne, Uninhabited, Cargo	A_{UC}	A_{UC} A_{UT} A_{UB}	Environmentally uncontrolled areas which cannot be inhabited by an aircrew during flight. Environmental extremes of pressure, temperature and shock may be severe. Examples include uninhabited areas of long mission aircraft such as the C130, C5, B52 and C141. This category also applies to uninhabited area of lower performance smaller aircraft such as the T38.
Airborne, Uninhabited, Fighter	A_{UF}	A_{UF} A_{UA}	Same as A_{UC} but installed on high performance aircraft such as fighters and interceptors. Examples include the F15, F16, F111 and A10 aircraft.
Airborne, Rotary Winged	A_{RW}	A_{RW}	Equipment installed on helicopters. Applies to both internally and externally mounted equipment such as laser designators, fire control systems, and communications equipment.
Space, Flight	S_F	S_F	Earth orbital. Approaches benign ground conditions. Vehicle neither under powered flight nor in atmospheric reentry; includes satellites and shuttles.

Table 3-2: Environmental Symbol and Description (cont'd)

Environment	π_E Symbol	Equivalent MIL-HDBK-217E, Notice 1 π_E Symbol	Description
Missile, Flight	M_F	M_{FF} M_{FA}	Conditions related to powered flight of air breathing missiles, cruise missiles, and missiles in unpowered free flight.
Missile, Launch	M_L	M_L U_{SL}	Severe conditions related to missile launch (air, ground and sea), space vehicle boost into orbit, and vehicle re-entry and landing by parachute. Also applies to solid rocket motor propulsion powered flight, and torpedo and missile launch from submarines.
Cannon, Launch	C_L	C_L	Extremely severe conditions related to cannon launching of 155 mm. and 5 inch guided projectiles. Conditions apply to the projectile from launch to target impact.

3.4.4 Part Failure Rate Models - Part failure rate models for microelectronic parts are significantly different from those for other parts and are presented entirely in Section 5.0. A typical example of the type of model used for most other part types is the following one for discrete semiconductors:

$$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_S \pi_C \pi_Q \pi_E$$

where:

λ_p is the part failure rate,

λ_b is the base failure rate usually expressed by a model relating the influence of electrical and temperature stresses on the part,

π_E and the other π factors modify the base failure rate for the category of environmental application and other parameters that affect the part reliability.

The π_E and π_Q factors are used in most all models and other π factors apply only to specific models. The applicability of π factors is identified in each section.

The base failure rate (λ_b) models are presented in each part section along with identification of the applicable model factors. Tables of calculated λ_b values are also provided for use in manual calculations. The model equations can, of course, be incorporated into computer programs for machine processing. The tabulated values of λ_b are cut off at the part ratings with regard to temperature and stress, hence, use of parts beyond these cut off points will overstress the part. The use of the λ_b models in a computer

program should take the part rating limits into account. The λ_b equations are mathematically continuous beyond the part ratings but such failure rate values are invalid in the overstressed regions.

All the part models include failure data from both catastrophic and permanent drift failures (e.g., a resistor permanently falling out of rated tolerance bounds) and are based upon a constant failure rate, except for motors which show an increasing failure rate over time. Failures associated with connection of parts into circuit assemblies are not included within the part failure rate models. Information on connection reliability is provided in Sections 16 and 17.

3.4.5 Thermal Aspects - The use of this prediction method requires the determination of the temperatures to which the parts are subjected. Since parts reliability is sensitive to temperature, the thermal analysis of any design should fairly accurately provide the ambient temperatures needed in using the part models. Of course, lower temperatures produce better reliability but also can produce increased penalties in terms of added loads on the environmental control system, unless achieved through improved thermal design of the equipment. The thermal analysis should be part of the design process and included in all the trade-off studies covering equipment performance, reliability, weight, volume, environmental control systems, etc. References 17 and 34 listed in Appendix C may be used as guides in determining component temperatures.

4.0 RELIABILITY ANALYSIS EVALUATION

Table 4-1 provides a general checklist to be used as a guide for evaluating a reliability prediction report. For completeness, the checklist includes categories for reliability modeling and allocation, which are sometimes delivered as part of a prediction report. It should be noted that the scope of any reliability analysis depends on the specific requirements called out in a statement-of-work (SOW) or system specification. The inclusion of this checklist is not intended to change the scope of these requirements.

Table 4-1: Reliability Analysis Checklist

Major Concerns	Comments
MODELS	
Are all functional elements included in the reliability block diagram /model?	System design drawings/diagrams must be reviewed to be sure that the reliability model/diagram agrees with the hardware.
Are all modes of operation considered in the math model?	Duty cycles, alternate paths, degraded conditions and redundant units must be defined and modeled.
Do the math model results show that the design achieves the reliability requirement?	Unit failure rates and redundancy equations are used from the detailed part predictions in the system math model (See MIL-STD-756, Reliability Prediction and Modeling).
ALLOCATION	
Are system reliability requirements allocated (subdivided) to useful levels?	Useful levels are defined as: equipment for subcontractors, assemblies for sub-subcontractors, circuit boards for designers.
Does the allocation process consider complexity, design flexibility, and safety margins?	Conservative values are needed to prevent reallocation at every design change.
PREDICTION	
Does the sum of the parts equal the value of the module or unit?	Many predictions neglect to include all the parts producing optimistic results (check for solder connections, connectors, circuit boards).
Are environmental conditions and part quality representative of the requirements?	Optimistic quality levels and favorable environmental conditions are often assumed causing optimistic results.
Are the circuit and part temperatures defined and do they represent the design?	Temperature is the biggest driver of part failure rates; low temperature assumptions will cause optimistic results.
Are equipment, assembly, subassembly and part reliability drivers identified?	Identification is needed so that corrective actions for reliability improvement can be considered.
Are alternate (Non MIL-HDBK-217) failure rates highlighted along with the rationale for their use?	Use of alternate failure rates, if deemed necessary, require submission of backup data to provide credence in the values.
Is the level of detail for the part failure rate models sufficient to reconstruct the result?	Each component type should be sampled and failure rates completely reconstructed for accuracy.
Are critical components such as VHSIC, Monolithic Microwave Integrated Circuits (MMIC), Application Specific Integrated Circuits (ASIC) or Hybrids highlighted?	Prediction methods for advanced technology parts should be carefully evaluated for impact on the module and system.

5.0 MICROCIRCUITS, INTRODUCTION

This section presents failure rate prediction models for the following ten major classes of microelectronic devices:

Section	
5.1	Monolithic Bipolar Digital and Linear Gate/Logic Array Devices
5.1	Monolithic MOS Digital and Linear Gate/Logic Array Devices
5.1	Monolithic Bipolar and MOS Digital Microprocessor Devices
5.2	Monolithic Bipolar and MOS Memory Devices
5.3	Very High Speed Integrated Circuit (VHSIC/VHSIC-Like and VLSI) CMOS Devices (> 60K Gates)
5.4	Monolithic GaAs Digital Devices
5.4	Monolithic GaAs MMIC
5.5	Hybrid Microcircuits
5.6	Surface Acoustic Wave Devices
5.7	Magnetic Bubble Memories

In the title description of each monolithic device type, Bipolar represents all TTL, ASTTL, DTL, ECL, CML, ALSTTL, HTTL, FTTL, F, LTTL, STTL, BiCMOS, LSTTL, IIL, I^3L and ISL devices. MOS represents all metal-oxide microcircuits, which includes NMOS, PMOS, CMOS and MNOS fabricated on various substrates such as sapphire, polycrystalline or single crystal silicon. The hybrid model is structured to accommodate all of the monolithic chip device types and various complexity levels.

Monolithic memory complexity factors are expressed in the number of bits in accordance with JEDEC STD 21A. This standard, which is used by all government and industry agencies that deal with microcircuit memories, states that memories of 1024 bits and greater shall be expressed as K bits, where 1K = 1024 bits. For example, a 16K memory has 16,384 bits, a 64K memory has 65,536 bits and a 1M memory has 1,048,576 bits. Exact numbers of bits are not used for memories of 1024 bits and greater.

For devices having both linear and digital functions not covered by MIL-M-38510 or MIL-I-38535, use the linear model. Line drivers and line receivers are considered linear devices. For linear devices not covered by MIL-M-38510 or MIL-I-38535, use the transistor count from the schematic diagram of the device to determine circuit complexity.

For digital devices not covered by MIL-M-38510 or MIL-I-38535, use the gate count as determined from the logic diagram. A J-K or R-S flip flop is equivalent to 6 gates when used as part of an LSI circuit. For the purpose of this Handbook, a gate is considered to be any one of the following functions; AND, OR, exclusive OR, NAND, NOR and inverter. When a logic diagram is unavailable, use device transistor count to determine gate count using the following expressions:

Technology	Gate Approximation
Bipolar	No. Gates = No. Transistors/3.0
CMOS	No. Gates = No. Transistors/4.0
All other MOS except CMOS	No. Gates = No. Transistors/3.0

5.0 MICROCIRCUITS, INTRODUCTION

A detailed form of the Section 5.3 VHSIC/VHSIC-Like model is included as Appendix B to allow more detailed trade-offs to be performed. Reference 30 should be consulted for more information about this model.

Reference 32 should be consulted for more information about the models appearing in Sections 5.1, 5.2, 5.4, 5.5, and 5.6. Reference 13 should be consulted for additional information on Section 5.7.

5.1 MICROCIRCUITS, GATE/LOGIC ARRAYS AND MICROPROCESSORS

DESCRIPTION

1. Bipolar Devices, Digital and Linear Gate/Logic Arrays
2. MOS Devices, Digital and Linear Gate/Logic Arrays
3. Field Programmable Logic Array (PLA) and Programmable Array Logic (PAL)
4. Microprocessors

$$\lambda_p = (C_1\pi_T + C_2\pi_E)\,\pi_Q\pi_L \text{ Failures}/10^6 \text{ Hours}$$

Bipolar Digital and Linear Gate/Logic Array Die Complexity Failure Rate - C_1

Digital		Linear		PLA/PAL	
No. Gates	C_1	No. Transistors	C_1	No. Gates	C_1
1 to 100	.0025	1 to 100	.010	Up to 200	.010
101 to 1,000	.0050	101 to 300	.020	201 to 1,000	.021
1,001 to 3,000	.010	301 to 1,000	.040	1,001 to 5,000	.042
3,001 to 10,000	.020	1,001 to 10,000	.060		
10,001 to 30,000	.040				
30,001 to 60,000	.080				

MOS Digital and Linear Gate/Logic Array Die Complexity Failure Rate - C_1*

Digital		Linear		PLA/PAL	
No. Gates	C_1	No. Transistors	C_1	No. Gates	C_1
1 to 100	.010	1 to 100	.010	Up to 500	.00085
101 to 1,000	.020	101 to 300	.020	501 to 1,000	.0017
1,001 to 3,000	.040	301 to 1,000	.040	2,001 to 5,000	.0034
3,001 to 10,000	.080	1,001 to 10,000	.060	5,001 to 20,000	.0068
10,001 to 30,000	.16				
30,001 to 60,000	.29				

*NOTE: For CMOS gate counts above 60,000 use the VHSIC/VHSIC-Like model in Section 5.3

Microprocessor
Die Complexity Failure Rate - C_1

No. Bits	Bipolar C_1	MOS C_1
Up to 8	.060	.14
Up to 16	.12	.28
Up to 32	.24	.56

All Other Model Parameters

Parameter	Refer to
π_T	Section 5.8
C_2	Section 5.9
π_E, π_Q, π_L	Section 5.10

5.2 MICROCIRCUITS, MEMORIES

DESCRIPTION

1. Read Only Memories (ROM)
2. Programmable Read Only Memories (PROM)
3. Ultraviolet Eraseable PROMs (UVEPROM)
4. "Flash," MNOS and Floating Gate Electrically Eraseable PROMs (EEPROM). Includes both floating gate tunnel oxide (FLOTOX) and textured polysilicon type EEPROMs
5. Static Random Access Memories (SRAM)
6. Dynamic Random Access Memories (DRAM)

$$\lambda_p = (C_1 \pi_T + C_2 \pi_E + \lambda_{cyc}) \pi_Q \pi_L \text{ Failures/}10^6 \text{ Hours}$$

Die Complexity Failure Rate - C_1

Memory Size, B (Bits)	MOS				Bipolar	
	ROM	PROM, UVEPROM, EEPROM, EAPROM	DRAM	SRAM (MOS & BiMOS)	ROM, PROM	SRAM
Up to 16K	.00065	.00085	.0013	.0078	.0094	.0052
16K < B ≤ 64K	.0013	.0017	.0025	.016	.019	.011
64K < B ≤ 256K	.0026	.0034	.0050	.031	.038	.021
256K < B ≤ 1M	.0052	.0068	.010	.062	.075	.042

A_1 Factor for λ_{cyc} Calculation

Total No. of Programming Cycles Over EEPROM Life, C	Flotox[1]	Textured-Poly[2]
Up to 100	.00070	.0097
100 < C ≤ 200	.0014	.014
200 < C ≤ 500	.0034	.023
500 < C ≤ 1K	.0068	.033
1K < C ≤ 3K	.020	.061
3K < C ≤ 7K	.049	.14
7K < C ≤ 15K	.10	.30
15K < C ≤ 20K	.14	.30
20K < C ≤ 30K	.20	.30
30K < C ≤ 100K	.68	.30
100K < C ≤ 200K	1.3	.30
200K < C ≤ 400K	2.7	.30
400K < C ≤ 500K	3.4	.30

1. $A_1 = 6.817 \times 10^{-6}$ (C)
2. No underlying equation for Textured-Poly.

A_2 Factor for λ_{cyc} Calculation

Total No. of Programming Cycles Over EEPROM Life, C	Textured-Poly A_2
Up to 300K	0
300K < C ≤ 400K	1.1
400K < C ≤ 500K	2.3

All Other Model Parameters

Parameter	Refer to
π_T	Section 5.8
C_2	Section 5.9
π_E, π_Q, π_L	Section 5.10
λ_{cyc} (EEPROMS only)	Page 5-5

$\lambda_{cyc} = 0$ For all other devices

EEPROM Read/Write Cycling Induced Failure Rate - λ_{cyc}

All Memory Devices Except Flotox and Textured-Poly EEPROMS	$\lambda_{cyc} = 0$
Flotox and Textured Poly EEPROMS	$\lambda_{cyc} = \left[A_1 B_1 + \frac{A_2 B_2}{\pi_Q} \right] \pi_{ECC}$

Model Factor	Flotox	Textured-Poly
A_1	Page 5-4	Page 5-4
B_1	Page 5-6	Page 5-6
A_2	$A_2 = 0$	Page 5-5
B_2	$B_2 = 0$	Page 5-6
π_Q	Section 5.10	Section 5.10
Error Correction Code (ECC) Options:		
1. No On-Chip ECC	$\pi_{ECC} = 1.0$	$\pi_{ECC} = 1.0$
2. On-Chip Hamming Code	$\pi_{ECC} = .72$	$\pi_{ECC} = .72$
3. Two-Needs-One Redundant Cell Approach	$\pi_{ECC} = .68$	$\pi_{ECC} = .68$

NOTES:

1. See Reference 24 for modeling off-chip error detection and correction schemes at the memory system level.
2. If EEPROM type is unknown, assume Flotox.
3. Error Correction Code Options: Some EEPROM manufacturers have incorporated on-chip error correction circuitry into their EEPROM devices. This is represented by the on-chip hamming code entry. Other manufacturers have taken a redundant cell approach which incorporates an extra storage transistor in every memory cell. This is represented by the two-needs-one redundant cell entry.
4. The A_1 and A_2 factors shown in Section 5.2 were developed based on an assumed system life of 10,000 operating hours. For EEPROMs used in systems with significantly longer or shorter expected lifetimes the A_1 and A_2 factors should be multiplied by:

$$\frac{10{,}000}{\text{System Lifetime Operating Hours}}$$

B_1 and B_2 Factors for λ_{cyc} Calculation

	Flotox[1] (B_1)					Textured-Poly[2] (B_1)					Textured-Poly[3] (B_2)				
Memory Size, B(Bits) → / T_J (°C)	4K	16K	64K	256K	1M	4K	16K	64K	256K	1M	4K	16K	64K	256K	1M
25	.27	0.55	1.1	2.2	4.3	.47	.66	.94	1.3	1.9	.54	0.76	1.1	1.5	2.1
30	.30	0.60	1.2	2.4	4.8	.50	.71	1.0	1.4	2.0	.50	0.71	1.0	1.4	2.0
35	.33	0.66	1.3	2.7	5.2	.54	.77	1.1	1.5	2.2	.47	0.67	.95	1.3	1.9
40	.36	0.72	1.4	2.9	5.7	.58	.82	1.2	1.6	2.3	.45	0.63	.89	1.3	1.8
45	.40	0.79	1.6	3.2	6.3	.62	.88	1.3	1.8	2.5	.42	0.59	.84	1.2	1.7
50	.43	0.86	1.7	3.4	6.8	.67	.94	1.3	1.9	2.7	.40	0.56	.80	1.1	1.6
55	.47	0.93	1.9	3.7	7.4	.71	1.0	1.4	2.0	2.8	.38	0.53	.75	1.1	1.5
60	.51	1.0	2.0	4.1	8.0	.76	1.1	1.5	2.1	3.0	.36	0.50	.72	1.0	1.4
65	.55	1.1	2.2	4.4	8.6	.81	1.1	1.6	2.3	3.2	.34	0.48	.68	.96	1.3
70	.59	1.2	2.4	4.7	9.3	.86	1.2	1.7	2.4	3.4	.32	0.46	.65	.91	1.3
75	.63	1.3	2.5	5.1	10	.91	1.3	1.8	2.6	3.6	.31	0.43	.62	.87	1.2
80	.68	1.4	2.7	5.4	11	.96	1.4	1.9	2.7	3.8	.29	0.41	.59	.83	1.2
85	.73	1.5	2.9	5.8	12	1.0	1.4	2.0	2.9	4.0	.28	0.40	.56	.79	1.1
90	.78	1.6	3.1	6.2	12	1.1	1.5	2.2	3.0	4.3	.27	0.38	.54	.75	1.1
95	.83	1.7	3.3	6.7	13	1.1	1.6	2.3	3.2	4.5	.26	0.36	.51	.72	1.0
100	.89	1.8	3.5	7.1	14	1.2	1.7	2.4	3.4	4.7	.25	0.35	.49	.69	.98
105	.94	1.9	3.8	7.5	15	1.3	1.8	2.5	3.5	5.0	.24	0.33	.47	.66	.94
110	1.0	2.0	4.0	8.0	16	1.3	1.9	2.6	3.7	5.2	.23	0.32	.45	.64	.90
115	1.1	2.1	4.2	8.5	17	1.4	1.9	2.8	3.9	5.5	.22	0.31	.44	.61	.87
120	1.1	2.2	4.5	9.0	18	1.4	2.0	2.9	4.1	5.7	.21	0.30	.42	.59	.83
125	1.2	2.4	4.7	9.5	19	1.5	2.1	3.0	4.2	6.0	.20	0.29	.41	.57	.80
130	1.3	2.5	5.0	10	20	1.6	2.2	3.2	4.4	6.2	.19	0.28	.39	.55	.77
135	1.3	2.6	5.3	11	21	1.6	2.3	3.3	4.6	6.5	.19	0.27	.38	.53	.75
140	1.4	2.8	5.6	11	22	1.7	2.4	3.4	4.8	6.8	.18	0.26	.36	.51	.72
145	1.5	2.9	5.8	12	23	1.8	2.5	3.6	5.0	7.1	.18	0.25	.35	.50	.70
150	1.5	3.1	6.1	12	24	1.8	2.6	3.7	5.2	7.4	.17	0.24	.34	.48	.68
155	1.6	3.2	6.4	13	25	1.9	2.7	3.9	5.4	7.6	.16	0.23	.33	.46	.65
160	1.7	3.4	6.7	14	27	2.0	2.8	4.0	5.6	7.9	.16	0.23	.32	.45	.63
165	1.8	3.5	7.1	14	28	2.1	2.9	4.2	5.8	8.2	.15	0.22	.31	.44	.62
170	1.8	3.7	7.4	15	29	2.1	3.0	4.3	6.1	8.5	.15	0.21	.30	.42	.60
175	1.9	3.9	7.7	15	31	2.2	3.1	4.5	6.3	8.8	.15	0.21	.29	.41	.58

1. $B_1 = \left(\frac{B}{16000}\right)^{.5}\left[\exp\left(\frac{-.15}{8.63 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{333}\right)\right)\right]$

2. $B_1 = \left(\frac{B}{64000}\right)^{.25}\left[\exp\left(\frac{-.12}{8.63 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{303}\right)\right)\right]$

3. $B_2 = \left(\frac{B}{64000}\right)^{.25}\left[\exp\left(\frac{.1}{8.63 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{303}\right)\right)\right]$

T_J = Worse Case Junction Temperature (°C). See Section 5.11 for T_J Determination

B = Number of bits. NOTE: 1K = 1024 bits

5.3 MICROCIRCUITS, VHSIC/VHSIC-LIKE AND VLSI CMOS

DESCRIPTION
CMOS greater than 60,000 gates

$$\lambda_p = \lambda_{BD}\pi_{MFG}\pi_T\pi_{CD} + \lambda_{BP}\pi_E\pi_Q\pi_{PT} + \lambda_{EOS} \text{ Failures/}10^6 \text{ Hours}$$

Die Base Failure Rate - λ_{BD}

Part Type	λ_{BD}
Logic and Custom	0.16
Gate Array	0.24

All Other Model Parameters

Parameter	Refer to
π_T	Section 5.8
π_E, π_Q	Section 5.10

Manufacturing Process Correction Factor - π_{MFG}

Manufacturing Process	π_{MFG}
QML or QPL	.55
Non QML or Non QPL	2.0

Package Type Correction Factor - π_{PT}

Package Type	π_{PT} Hermetic	π_{PT} Nonhermetic
DIP	1.0	1.3
Pin Grid Array	2.2	2.9
Chip Carrier (Surface Mount Technology)	4.7	6.1

Die Complexity Correction Factor - π_{CD}

Feature Size (Microns)	Die Area (cm^2) A ≤ .4	.4 < A ≤ .7	.7 < A ≤ 1.0	1.0 < A ≤ 2.0	2.0 < A ≤ 3.0
.80	8.0	14	19	38	58
1.00	5.2	8.9	13	25	37
1.25	3.5	5.8	8.2	16	24

$$\pi_{CD} = \left(\left(\frac{A}{.21}\right)\left(\frac{2}{X_s}\right)^2 (.64)\right) + .36$$

A = Total Scribed Chip Die Area in cm^2 X_s = Feature Size (microns)

Die Area Conversion: $cm^2 = MIL^2 \div 155{,}000$

Package Base Failure Rate - λ_{BP}

Number of Pins	λ_{BP}
24	.0026
28	.0027
40	.0029
44	.0030
48	.0030
52	.0031
64	.0033
84	.0036
120	.0043
124	.0043
144	.0047
220	.0060

$$\lambda_{BP} = .0022 + \left((1.72 \times 10^{-5})(NP)\right)$$

NP = Number of Package Pins

Electrical Overstress Failure Rate - λ_{EOS}

V_{TH} (ESD Susceptibility (Volts))*	λ_{EOS}
0 - 1000	.065
> 1000 - 2000	.053
> 2000 - 4000	.044
> 4000 - 16000	.029
> 16000	.0027

$$\lambda_{EOS} = (-\ln(1 - .00057\exp(-.0002\,V_{TH}))) / .00876$$

V_{TH} = ESD Susceptibility (volts)

* Voltage ranges which will cause the part to fail. If unknown, use 0 - 1000 volts.

5.4 MICROCIRCUITS, GaAs MMIC AND DIGITAL DEVICES

DESCRIPTION
Gallium Arsenide Microwave Monolithic Integrated Circuit (GaAs MMIC) and GaAs Digital Integrated Circuits using MESFET Transistors and Gold Based Metallization

$$\lambda_p = [\, C_1 \pi_T \pi_A + C_2 \pi_E \,] \pi_L \pi_Q \quad \text{Failures}/10^6 \text{ Hours}$$

MMIC: Die Complexity Failure Rates - C_1

Complexity (No. of Elements)	C_1
1 to 100	4.5
101 to 1000	7.2
1. C_1 accounts for the following active elements: transistors, diodes.	

Digital: Die Complexity Failure Rates - C_1

Complexity (No. of Elements)	C_1
1 to 1000	25
1,001 to 10,000	51
1. C_1 accounts for the following active elements: transistors, diodes.	

Device Application Factor - π_A

Application	π_A
MMIC Devices	
Low Noise & Low Power (≤ 100 mW)	1.0
Driver & High Power (> 100 mW)	3.0
Unknown	3.0
Digital Devices	
All Digital Applications	1.0

All Other Model Parameters

Parameter	Refer to
π_T	Section 5.8
C_2	Section 5.9
π_E, π_L, π_Q	Section 5.10

5.5 MICROCIRCUITS, HYBRIDS

DESCRIPTION
Hybrid Microcircuits

$$\lambda_p = [\Sigma N_C \lambda_C] (1 + .2 \pi_E) \pi_F \pi_Q \pi_L \text{ Failures/}10^6 \text{ Hours}$$

N_C = Number of Each Particular Component
λ_C = Failure Rate of Each Particular Component

The general procedure for developing an overall hybrid failure rate is to calculate an individual failure rate for each component type used in the hybrid and then sum them. This summation is then modified to account for the overall hybrid function (π_F), screening level (π_Q), and maturity (π_L). The hybrid package failure rate is a function of the active component failure modified by the environmental factor (i.e., $(1 + .2 \pi_E)$). Only the component types listed in the following table are considered to contribute significantly to the overall failure rate of most hybrids. All other component types (e.g., resistors, inductors, etc.) are considered to contribute insignificantly to the overall hybrid failure rate, and are assumed to have a failure rate of zero. This simplification is valid for most hybrids; however, if the hybrid consists of mostly passive components then a failure rate should be calculated for these devices. If factoring in other component types, assume $\pi_Q = 1$, $\pi_E = 1$ and T_A = Hybrid Case Temperature for these calculations.

Determination of λ_C

Determine λ_C for These Component Types	Handbook Section	Make These Assumptions When Determining λ_C
Microcircuits	5	$C_2 = 0$, $\pi_Q = 1$, $\pi_L = 1$, T_J as Determined from Section 5.12, $\lambda_{BP} = 0$ (for VHSIC).
Discrete Semiconductors	6	$\pi_Q = 1$, T_J as Determined from Section 6.14, $\pi_E = 1$.
Capacitors	10	$\pi_Q = 1$, T_A = Hybrid Case Temperature, $\pi_E = 1$.

NOTE: If maximum rated stress for a die is unknown, assume the same as for a discretely package die of the same type. If the same die has several ratings based on the discrete packaged type, assume the lowest rating. Power rating used should be based on case temperature for discrete semiconductors.

Circuit Function Factor - π_F

Circuit Type	π_F
Digital	1.0
Video, 10 MHz < f < 1 GHz	1.2
Microwave, f > 1 GHz	2.6
Linear, f < 10 MHz	5.8
Power	21

All Other Hybrid Model Parameters

π_L, π_Q, π_E	Refer to Section 5.10

5.6 MICROCIRCUITS, SAW DEVICES

DESCRIPTION
Surface Acoustic Wave Devices

$$\lambda_p = 2.1\ \pi_Q\ \pi_E \text{ Failures}/10^6 \text{ Hours}$$

Quality Factor - π_Q

Screening Level	π_Q
10 Temperature Cycles (-55°C to +125°C) with end point electrical tests at temperature extremes.	.10
None beyond best commerical practices.	1.0

Environmental Factor - π_E

Environment	π_E
G_B	.5
G_F	2.0
G_M	4.0
N_S	4.0
N_U	6.0
A_{IC}	4.0
A_{IF}	5.0
A_{UC}	5.0
A_{UF}	8.0
A_{RW}	8.0
S_F	.50
M_F	5.0
M_L	12
C_L	220

5.7 MICROCIRCUITS, MAGNETIC BUBBLE MEMORIES

The magnetic bubble memory device in its present form is a non-hermetic assembly consisting of the following two major structural segments:

1. A basic bubble chip or die consisting of memory or a storage area (e.g., an array of minor loops), and required control and detection elements (e.g., generators, various gates and detectors).

2. A magnetic structure to provide controlled magnetic fields consisting of permanent magnets, coils, and a housing.

These two structural segments of the device are interconnected by a mechanical substrate and lead frame. The interconnect substrate in the present technology is normally a printed circuit board. It should be noted that this model does not include external support microelectronic devices required for magnetic bubble memory operation. The model is based on Reference 33. The general form of the failure rate model is:

$$\lambda_p = \lambda_1 + \lambda_2 \quad \text{Failures/}10^6 \text{ Hours}$$

where:

λ_1 = Failure Rate of the Control and Detection Structure

$$\lambda_1 = \pi_Q [N_C C_{11} \pi_{T1} \pi_W + (N_C C_{21} + C_2)\pi_E]\pi_D \pi_L$$

λ_2 = Failure Rate of the Memory Storage Area

$$\lambda_2 = \pi_Q N_C (C_{12} \pi_{T2} + C_{22} \pi_E)\pi_L$$

Chips Per Package - N_C

N_C = Number of Bubble Chips per Packaged Device

Temperature Factor – π_T

$$\pi_T = (.1) \exp\left[\frac{-Ea}{8.63 \times 10^{-5}} \left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right]$$

Use:

E_a = .8 to Calculate π_{T1}

E_a = .55 to Calculate π_{T2}

T_J = Junction Temperature (°C), $25 \le T_J \le 175$

T_J = T_{CASE} + 10°C

Device Complexity Failure Rates for Control and Detection Structure - C_{11} and C_{21}

$C_{11} = .00095(N_1)^{.40}$

$C_{21} = .0001(N_1)^{.226}$

N_1 = Number of Dissipative Elements on a Chip (gates, detectors, generators, etc.), $N_1 \le 1000$

5.7 MICROCIRCUIT, MAGNETIC BUBBLE MEMORIES

Write Duty Cycle Factor - π_W

$$\pi_W = \frac{10D}{(R/W)^{.3}}$$

$\pi_W = 1$ for $D \leq .3$ or $R/W \geq 2154$

$$D = \frac{\text{Avg. Device Data Rate}}{\text{Mfg. Max. Rated Data Rate}} \leq 1$$

R/W = No. of Reads per Write

NOTE:
For seed-bubble generators, divide π_W by 4, or use 1, whichever is greater.

Duty Cycle Factor - π_D

$$\pi_D = .9D + .1$$

$$D = \frac{\text{Avg. Device Data Rate}}{\text{Mfg. Max. Rated Data Rate}} \leq 1$$

Device Complexity Failure Rates for Memory Storage Structure - C_{12} and C_{22}

$$C_{12} = .00007(N_2)^{.3}$$

$$C_{22} = .00001(N_2)^{.3}$$

N_2 = Number of Bits, $N_2 \leq 9 \times 10^6$

All Other Model Parameters

Parameter	Section
C_2	5.9
π_E, π_Q, π_L	5.10

Temperature Factor For All Microcircuits - π_T

	TTL, ASTTL, CML, HTTL, FTTL, DTL, ECL, ALSTTL	F, LTTL, STTL	BiCMOS, LSTTL	III, I^2L, ISL	Digital MOS, VHSIC CMOS	Linear (Bipolar & MOS)	Memories (Bipolar & MOS), MNOS	GaAs MMIC Active Devices, π_{TA}	GaAs Digital Active Devices, π_{TA}
Ea(eV) → T_J (°C)	.4	.45	.5	.6	.35	.65	.6	1.5	1.4
25	.10	.10	.10	.10	.10	.10	.10	3.20E-09	1.00E-08
30	.13	.13	.14	.15	.13	.15	.15	8.40E-09	2.50E-08
35	.17	.18	.19	.21	.16	.23	.21	2.10E-08	5.90E-08
40	.21	.23	.25	.31	.19	.34	.31	5.20E-08	1.40E-07
45	.27	.3	.34	.43	.24	.49	.43	1.30E-07	3.10E-07
50	.33	.39	.45	.61	.29	.71	.61	2.90E-07	6.80E-07
55	.42	.50	.59	.85	.35	1.0	.85	6.70E-07	1.50E-06
60	.51	.63	.77	1.2	.42	1.4	1.2	1.50E-06	3.10E-06
65	.63	.80	1.0	1.6	.50	2.0	1.6	3.20E-06	6.40E-06
70	.77	1.0	1.3	2.1	.60	2.8	2.1	6.80E-06	1.30E-05
75	.94	1.2	1.6	2.9	.71	3.8	2.9	1.40E-05	2.50E-05
80	1.1	1.5	2.1	3.8	.84	5.2	3.8	2.90E-05	4.90E-05
85	1.4	1.9	2.6	5.0	.98	7.0	5.0	5.70E-05	9.40E-05
90	1.6	2.3	3.3	6.6	1.1	9.3	6.6	1.10E-04	1.70E-04
95	1.9	2.8	4.1	8.5	1.3	12	8.5	2.10E-04	3.20E-04
100	2.3	3.4	5.0	11	1.5	16	11	4.00E-04	5.80E-04
105	2.7	4.1	6.2	14	1.8	21	14	7.50E-04	1.00E-03
110	3.2	4.9	7.5	18	2.1	28	18	1.40E-03	1.80E-03
115	3.7	5.8	9.2	23	2.4	35	23	2.40E-03	3.10E-03
120	4.3	6.9	11	28	2.7	45	28	4.30E-03	5.30E-03
125	5	8.2	13	35	3.1	58	35	7.50E-03	9.00E-03
130	5.8	9.6	16	44	3.5	73	44	1.30E-02	1.50E-02
135	6.7	11	19	54	3.9	92	54	2.20E-02	2.40E-02
140	7.7	13	23	67	4.4	120	67	3.70E-02	3.90E-02
145	8.8	15	27	82	5.0	140	82	6.10E-02	6.30E-02
150	10	18	32	100	5.6	180	100	1.00E-01	1.00E-01
155	11	20	37	120	6.3	220	120	1.60E-01	1.60E-01
160	13	24	43	150	7.0	270	150	2.60E-01	2.40E-01
165	15	27	50	180	7.8	330	180	4.10E-01	3.70E-01
170	18	31	59	210	8.7	400	210	6.40E-01	5.70E-01
175	18	35	68	250	9.6	480	250	9.90E-01	8.50E-01

$$\pi_T = .1 \exp\left(\frac{-Ea}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{298}\right)\right) \text{ Silicon Devices} \qquad \pi_T = .1 \exp\left(\frac{-Ea}{8.617 \times 10^{-5}}\left(\frac{1}{T_J + 273} - \frac{1}{423}\right)\right) \text{ GaAs Devices}$$

E_a = Effective Activation Energy (eV) (Shown Above)

T_J = Worse Case Junction Temperature (Silicon Devices) or Average Active Device Channel Temperature (GaAs Devices). See Section 5.11 (or Section 5.12 for Hybrids) for T_J Determination.

NOTES:
1. $T_J = T_C + P\theta_{JC}$
   T_C = Case Temperature (°C)
   P = Device Power Dissipation (W)
   θ_{JC} = Junction to Case Thermal Resistance (°C/W)
   θ_{JC} should be obtained from the device manufacturer, MIL-M-38510, or from the default values shown in Section 5.11 for the closest equivalent device.
2. Use Digital MOS column for HC, HCT, AC, ACT, C and FCT technologies.
3. Table entries should be considered valid only up to the rated temperature of the component under consideration.

5.9 MICROCIRCUITS, C_2 TABLE FOR ALL

Package Failure Rate for all Microcircuits - C_2

Package Type					
Number of Functional Pins, N_p	Hermetic: DIPs w/Solder or Weld Seal, Pin Grid Array (PGA)[1], SMT (Leaded and Nonleaded)	DIPs with Glass Seal[2]	Flatpacks with Axial Leads on 50 Mil Centers[3]	Cans[4]	Nonhermetic: DIPs, PGA, SMT (Leaded and Nonleaded)[5]
3	.00092	.00047	.00022	.00027	.0012
4	.0013	.00073	.00037	.00049	.0016
6	.0019	.0013	.00078	.0011	.0025
8	.0026	.0021	.0013	.0020	.0034
10	.0034	.0029	.0020	.0031	.0043
12	.0041	.0038	.0028	.0044	.0053
14	.0048	.0048	.0037	.0060	.0062
16	.0056	.0059	.0047	.0079	.0072
18	.0064	.0071	.0058		.0082
22	.0079	.0096	.0083		.010
24	.0087	.011	.0098		.011
28	.010	.014			.013
36	.013	.020			.017
40	.015	.024			.019
64	.025	.048			.032
80	.032				.041
128	.053				.068
180	.076				.098
224	.097				.12

1. $C_2 = 2.8 \times 10^{-4} (N_p)^{1.08}$
2. $C_2 = 9.0 \times 10^{-5} (N_p)^{1.51}$
3. $C_2 = 3.0 \times 10^{-5} (N_p)^{1.82}$
4. $C_2 = 3.0 \times 10^{-5} (N_p)^{2.01}$
5. $C_2 = 3.6 \times 10^{-4} (N_p)^{1.08}$

NOTES:

1. SMT: Surface Mount Technology
2. DIP: Dual In-Line Package
3. If DIP Seal type is unknown, assume glass
4. The package failure rate (C_2) accounts for failures associated only with the package itself. Failures associated with mounting the package to a circuit board are accounted for in Section 16, Interconnection Assemblies.

5.10 MICROCIRCUITS, π_E, λ_L AND π_Q TABLES FOR ALL

Environment Factor - π_E

Environment	π_E
G_B	.50
G_F	2.0
G_M	4.0
N_S	4.0
N_U	6.0
A_{IC}	4.0
A_{IF}	5.0
A_{UC}	5.0
A_{UF}	8.0
A_{RW}	8.0
S_F	.50
M_F	5.0
M_L	12
C_L	220

Learning Factor - π_L

Years in Production, Y	π_L
≤ .1	2.0
.5	1.8
1.0	1.5
1.5	1.2
≥ 2.0	1.0

$\pi_L = .01 \exp(5.35 - .35Y)$

Y = Years generic device type has been in production

Quality Factors - π_Q

Description	π_Q
Class S Categories: 1. Procured in full accordance with MIL-M-38510, Class S requirements. 2. Procured in full accordance with MIL-I-38535 and Appendix B thereto (Class U). 3. Hybrids: (Procured to Class S requirements (Quality Level K) of MIL-H-38534.	.25
Class B Categories: 1. Procured in full accordance with MIL-M-38510, Class B requirements. 2. Procured in full accordance with MIL-I-38535, (Class Q). 3. Hybrids: Procured to Class B requirements (Quality Level H) of MIL-H-38534.	1.0
Class B-1 Category: Fully compliant with all requirements of paragraph 1.2.1 of MIL-STD-883 and procured to a MIL drawing, DESC drawing or other government approved documentation. (Does not include hybrids). For hybrids use custom screening section below.	2.0

5.10 MICROCIRCUITS, π_E, π_L AND π_Q TABLES FOR ALL

Quality Factors (cont'd): π_Q Calculation for Custom Screening Programs

Group	MIL-STD-883 Screen/Test (Note 3)	Point Valuation	
1*	TM 1010 (Temperature Cycle, Cond B Minimum) and TM 2001 (Constant Acceleration, Cond B Minimum) and TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual)	50	
2*	TM 1010 (Temperature Cycle, Cond B Minimum) or TM 2001 (Constant Acceleration, Cond B Minimum) TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temp Extremes) and TM 1014 (Seal Test, Cond A, B, or C) and TM 2009 (External Visual)	37	
3	Pre-Burn in Electricals TM 1015 (Burn-in B-Level/S-Level) and TM 5004 (or 5008 for Hybrids) (Post Burn-in Electricals @ Temp Extremes)	30 36	(B Level) (S Level)
4*	TM 2020 Pind (Particle Impact Noise Detection)	11	
5	TM 5004 (or 5008 for Hybrids) (Final Electricals @ Temperature Extremes)	11	(Note 1)
6	TM 2010/17 (Internal Visual)	7	
7*	TM 1014 (Seal Test, Cond A, B, or C)	7	(Note 2)
8	TM 2012 (Radiography)	7	
9	TM 2009 (External Visual)	7	(Note 2)
10	TM 5007/5013 (GaAs) (Wafer Acceptance)	1	
11	TM 2023 (Non-Destructive Bond Pull)	1	

$$\pi_Q = 2 + \frac{87}{\Sigma \text{ Point Valuations}}$$

*NOT APPROPRIATE FOR PLASTIC PARTS.

NOTES:
1. Point valuation only assigned if used independent of Groups 1, 2 or 3.
2. Point valuation only assigned if used independent of Groups 1 or 2.
3. Sequencing of tests within groups 1, 2 and 3 must be followed.
4. TM refers to the MIL-STD-883 Test Method.
5. Nonhermetic parts should be used only in controlled environments (i.e., G_B and other temperature/humidity controlled environments).

EXAMPLES:

1. Mfg. performs Group 1 test and Class B burn-in: $\pi_Q = 2 + \frac{87}{50+30} = 3.1$
2. Mfg. performs internal visual test, seal test and final electrical test: $\pi_Q = 2 + \frac{87}{7+7+11} = 5.5$

Other Commercial or Unknown Screening Levels	$\pi_Q = 10$

5.11 MICROCIRCUITS, T_J DETERMINATION, (ALL EXCEPT HYBRIDS)

Ideally, device case temperatures should be determined from a detailed thermal analysis of the equipment. Device junction temperature is then calculated with the following relationship:

$$T_J = T_C + \theta_{JC}P$$

T_J = Worst Case Junction Temperature (°C).

T_C = Case Temperature (°C). If not available, use the following default table.

Default Case Temperature (T_C) for all Environments

Environment	G_B	G_F	G_M	N_S	N_U	A_{IC}	A_{IF}	A_{UC}	A_{UF}	A_{RW}	S_F	M_F	M_L	C_L
T_C (°C)	35	45	50	45	50	60	60	75	75	60	35	50	60	45

θ_{JC} = Junction-to-case thermal resistance (°C/watt) for a device soldered into a printed circuit board. If θ_{JC} is not available, use a value contained in a specification for the closest equivalent device or use the following table.

Package Type (Ceramic Only)	Die Area > 14,400 mil^2 θ_{JC} (°C/W)	Die Area ≤ 14,400 mil^2 θ_{JC} (°C/W)
Dual-In-Line	11	28
Flat Package	10	22
Chip Carrier	10	20
Pin Grid Array	10	20
Can	–	70

P = The maximum power dissipation realized in a system application. If the applied power is not available, use the maximum power dissipation from the specification for the closest equivalent device.

5.12 MICROCIRCUITS, T_J DETERMINATION, (FOR HYBRIDS)

This section describes a method for estimating junction temperature (T_J) for integrated circuit dice mounted in a hybrid package. A hybrid is normally made up of one or more substrate assemblies mounted within a sealed package. Each substrate assembly consists of active and passive chips with thick or thin film metallization mounted on the substrate, which in turn may have multiple layers of metallization and dielectric on the surface. Figure 5-1 is a cross-sectional view of a hybrid with a single multi-layered substrate. The layers within the hybrid are made up of various materials with different thermal characteristics. The table following Figure 5-1 provides a list of commonly used hybrid materials with typical thicknesses and corresponding thermal conductivities (K). If the hybrid internal structure cannot be determined, use the following default values for the temperature rise from case to junction: microcircuits, 10°C; transistors, 25°C; diodes, 20°C. Assume capacitors are at T_C.

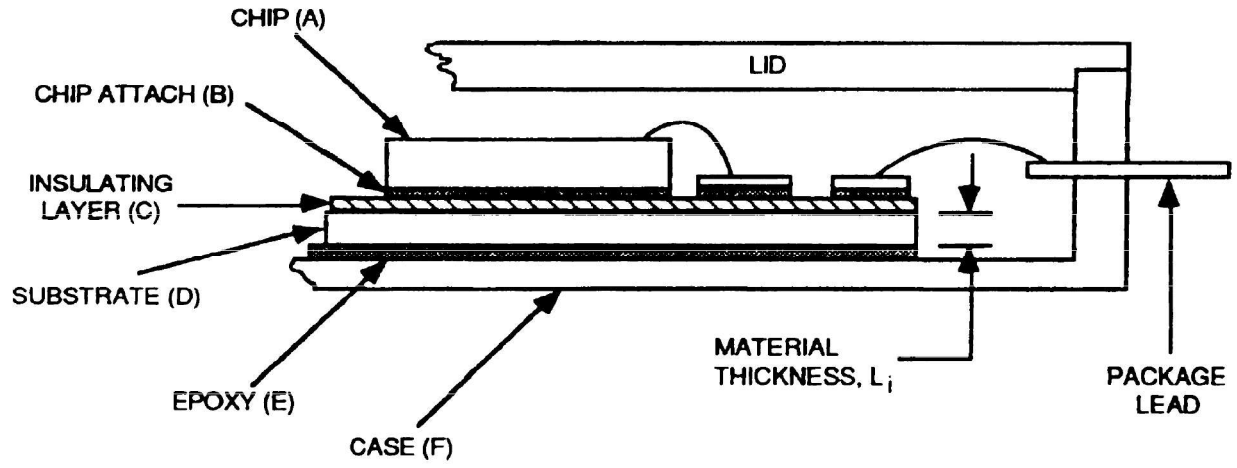


Figure 5-1: Cross-sectional View of a Hybrid with a Single Multi-Layered Substrate

5.12 MICROCIRCUITS, T_J DETERMINATION, (FOR HYBRIDS)

Typical Hybrid Characteristics

Material	Typical Usage	Typical Thickness, L_i (in.)	Feature From Figure 5-1	Thermal Conductivity, K_i $\left(\frac{W/in^2}{°C/in}\right)$	$\left(\frac{1}{K_i}\right)(L_i)$ $\left(in^2\ °C/W\right)$
Silicon	Chip Device	0.010	A	2.20	.0045
GaAs	Chip Device	0.0070	A	.76	.0092
Au Eutectic	Chip Attach	0.0001	B	6.9	.000014
Solder	Chip/Substrate Attach	0.0030	B/E	1.3	.0023
Epoxy (Dielectric)	Chip/Substrate Attach	0.0035	B/E	.0060	.58
Epoxy (Conductive)	Chip Attach	0.0035	B	.15	.023
Thick Film Dielectric	Glass Insulating Layer	0.0030	C	.66	.0045
Alumina	Substrate, MHP	0.025	D	.64	.039
Beryllium Oxide	Substrate, PHP	0.025	D	6.6	.0038
Kovar	Case, MHP	0.020	F	.42	.048
Aluminum	Case, MHP	0.020	F	4.6	.0043
Copper	Case, PHP	0.020	F	9.9	.0020

NOTE: MHP: Multichip Hybrid Package, PHP: Power Hybrid Package (Pwr: ≥ 2W, Typically)

$$\theta_{JC} = \frac{\sum_{i=1}^{n} \left(\frac{1}{K_i}\right)\left(L_i\right)}{A}$$

n = Number of Material Layers

K_i = Thermal Conductivity of ith Material $\left(\frac{W/in^2}{°C/in}\right)$ (User Provided or From Table)

L_i = Thickness of ith Material (in) (User Provided or From Table)

A = Die Area (in^2). If Die Area cannot be readily determined, estimate as follows:
$A = [\ .00278\ (\text{No. of Die Active Wire Terminals}) + .0417]^2$

Estimate T_J as Follows:

$$T_J = T_C + .9\ (\theta_{JC})\ (P_D)$$

T_C = Hybrid Case Temperature (°C). If unknown, use the T_C Default Table shown in Section 5.11.

θ_{JC} = Junction-to-Case Thermal Resistance (°C/W) (As determined above)

P_D = Die Power Dissipation (W)

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Example 1: CMOS Digital Gate Array

Given: A CMOS digital timing chip (4046) in an airborne inhabited cargo application, case temperature 48°C, 75mW power dissipation. The device is procured with normal manufacturer's screening consisting of temperature cycling, constant acceleration, electrical testing, seal test and external visual inspection, in the sequence given. The component manufacturer also performs a B-level burn-in followed by electrical testing. All screens and tests are performed to the applicable MIL-STD-883 screening method. The package is a 24 pin ceramic DIP with a glass seal. The device has been manufactured for several years and has 1000 transistors.

$$\lambda_p = (C_1\pi_T + C_2\pi_E)\,\pi_Q\pi_L \qquad \text{Section 5.1}$$

$C_1 = .020$ — 1000 Transistors = 250 Gates, MOS C_1 Table, Digital Column

$\pi_T = .29$ — Determine T_J from Section 5.11
$T_J = 48°C + (28°C/W)(.075W) = 50°C$
Determine π_T from Section 5.8, Digital MOS Column.

$C_2 = .011$ — Section 5.9

$\pi_E = 4.0$ — Section 5.10

$\pi_Q = 3.1$ — Section 5.10

Group 1 Tests	50 Points
Group 3 Tests (B-level)	30 Points
TOTAL	80 Points

$$\pi_Q = 2 + \frac{87}{80} = 3.1$$

$\pi_L = 1$ — Section 5.10

$$\lambda_p = [\,(.020)(.29) + (.011)\,(4)\,]\,(3.1)(1) = .15 \text{ Failure}/10^6 \text{ Hours}$$

Example 2: EEPROM

Given: A 128K Flotox EEPROM that is expected to have a T_J of 80°C and experience 10,000 read/write cycles over the life of the system. The part is procured to all requirements of Paragraph 1.2.1, MIL-STD-883, Class B screening level requirements and has been in production for three years. It is packaged in a 28 pin DIP with a glass seal and will be used in an airborne uninhabited cargo application.

$$\pi_p = (C_1\,\pi_T + C_2\,\pi_E + \lambda_{cyc})\,\pi_Q\,\pi_L \qquad \text{Section 5.2}$$

$C_1 = .0034$ — Section 5.2

$\pi_T = 3.8$ — Section 5.8

$C_2 = .014$ — Section 5.9

π_E	= 5.0	Section 5.10
π_Q	= 2.0	Section 5.10
π_L	= 1.0	Section 5.10
λ_{cyc}	= .38	Section 5.2:

$$\lambda_{cyc} = \left[A_1 B_1 + \frac{A_2 B_2}{\pi_Q} \right] \pi_{ECC}$$

$A_2 = B_2 = 0$ for Flotox

Assume No ECC, $\pi_{ECC} = 1$

$A_1 = .1$, $7K \leq C \leq 15K$ Entry

$B_1 = 3.8$ (Use Equation 1 at bottom of B_1 and B_2 Table)

$\lambda_{cyc} = A_1 B_1 = (.1)(3.8) = .38$

$$\lambda_p = [\,(.0034)(3.8) + (.014)(5.0) + .38\,]\,(2.0)(1) = .93 \text{ Failures}/10^6 \text{ Hours}$$

Example 3: GaAs MMIC

Given: A MA4GM212 Single Pole Double Throw Switch, DC - 12 GHz, 4 transistors, 4 inductors, 8 resistors, maximum input $P_D = 30$ dbm, 16 pin hermetic flatpack, maximum $T_{CH} = 145°C$ in a ground benign environment. The part has been manufactured for 1 year and is screened to Paragraph 1.2.1 of MIL-STD-883, Class B equivalent screen.

$$\lambda_p = [\,C_1\pi_T\pi_A + C_2\pi_E\,]\,\pi_L\pi_Q \qquad \text{Section 5.4}$$

C_1	= 4.5	Section 5.4, MMIC Table, 4 Active Elements (See Footnote to Table)
π_T	= .061	Section 5.8, $T_J = T_{CH} = 145°C$
π_A	= 3.0	Section 5.4, Unknown Application
C_2	= .0047	Section 5.9
π_E	= .50	Section 5.10
π_L	= 1.5	Section 5.10
π_Q	= 2.0	Section 5.10

$$\lambda_p = \left[(4.5)(.061)(3.0) + (.0047)(.5)\right](1.5)(2.0) = 2.5 \text{ Failures}/10^6 \text{ Hours}$$

NOTE: The passive elements are assumed to contribute negligibly to the overall device failure rate.

Example 4: Hybrid

Given: A linear multichip hybrid driver in a hermetically sealed Kovar package. The substrate is alumina and there are two thick film dielectric layers. The die and substrate attach materials are conductive epoxy and solder, respectively. The application environment is naval unsheltered, 65°C case temperature and the device has been in production for over two years. The device is

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screened to MIL-STD-883, Method 5008, in accordance with Table VIII, Class B requirements. The hybrid contains the following components:

Active Components:	1	-	LM106 Bipolar Comparator/Buffer Die (13 Transistors)
	1	-	LM741A Bipolar Operational Amplifier Die (24 Transistors)
	2	-	Si NPN Transistor
	2	-	Si PNP Transistor
	2	-	Si General Purpose Diodes
Passive Components:	2	-	Ceramic Chip Capacitors
	17	-	Thick Film Resistors

$$\lambda_p = [\Sigma N_C \lambda_C] (1 + .2\pi_E) \pi_F \pi_Q \pi_L \quad \text{Section 5.5}$$

1. Estimate Active Device Junction Temperatures

If limited information is available on the specific hybrid materials and construction characteristics the default case-to-junction temperature rises shown in the introduction to Section 5.12 can be used. When detailed information becomes available the following Section 5.12 procedure should be used to determine the junction-to-case (θ_{JC}) thermal resistance and T_J values for each component.

$$\theta_{JC} = \frac{\sum_{i=1}^{n} \left(\frac{1}{K_i}\right)(L_i)}{A} \quad \text{(Equation 1)}$$

Layer	Figure 5-1 Feature		$\left(\frac{1}{K_i}\right)(L_i)$ $\left(in^2\ °C/W\right)$
Silicon Chip	A		.0045
Conductive Epoxy	B		.023
Two Dielectric Layers	C	(2)(.0045) =	.009
Alumina Substrate	D		.039
Solder Substrate Attachment	E		.0023
Kovar Case	F		.048
		$\Sigma\left(\frac{1}{K_i}\right)(L_i)$ =	.1258

$$A = \text{Die Area} = [.00278 (\text{No. Die Active Wire Terminals}) + .0417]^2 \quad \text{(Equation 2)}$$

$$T_J = T_C + \theta_{JC} P_D \quad \text{(Equation 3)}$$

	LM106	LM741A	Si NPN	Si PNP	Si Diode	Source
No. of Pins	8	14	3	3	2	Vendor Spec. Sheet
Power Dissipation, P_D (W)	.33	.35	.6	.6	.42	Circuit Analysis
Area of Chip (in.^{2})	.0041	.0065	.0025	.0025	.0022	Equ. 2 Above
θ_{JC} (°C/W)	30.8	19.4	50.3	50.3	56.3	Equ. 1 Above
T_J (°C)	75	72	95	95	89	Equ. 3 Above

2. Calculate Failure Rates for Each Component:

A) LM106 Die, 13 Transistors (from Vendor Spec. Sheet)

$\lambda_p = [\, C_1 \pi_T + C_2 \pi_E \,] \pi_Q \pi_L$ Section 5.1

Because $C_2 = 0$;

$\lambda_p = C_1 \pi_T \pi_Q \pi_L$ π_T: Section 5.8; π_Q, π_L Default to 1.0

$= (.01)(3.8)(1)(1) = .038$ Failures/10^6 Hours

B) LM741 Die, 23 Transistors. Use Same Procedure as Above.

$\lambda_p = C_1 \pi_T \pi_Q \pi_L = (.01)(3.1)(1)(1) = .031$ Failures/10^6 Hours

C) Silicon NPN Transistor, Rated Power = 5W (From Vendor Spec. Sheet), $V_{CE}/V_{CEO} = .6$, Linear Application

$\lambda_p = \lambda_b \pi_T \pi_A \pi_R \pi_S \pi_Q \pi_E$ Section 6.3; π_Q, π_E Default to 1.0

$= (.00074)(3.9)(1.5)(1.8)(.29)(1)(1)$

$= .0023$ Failures/10^6 Hours

D) Silicon PNP Transistor, Same as C.

$\lambda_p = .0023$ Failures/10^6 Hours

E) Silicon General Purpose Diode (Analog), Voltage Stress = 60%, Metallurgically Bonded Construction.

$\lambda_p = \lambda_b \pi_T \pi_S \pi_C \pi_Q \pi_E$ Section 6.1; π_Q, π_E Default to 1.0

$= (.0038)(6.3)(.29)(1)(1)(1)$

$= .0069$ Failures/10^6 Hours

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F) Ceramic Chip Capacitor, Voltage Stress = 50%,
$T_A = T_{CASE}$ for the Hybrid, 1340 pF, 125°C Rated Temp.

$\lambda_p = \lambda_b \pi_{CV} \pi_Q \pi_E$ Section 10.11; π_Q, π_E Default to 1.0
$= (.0028)(1.4)(1)(1)$
$= .0039$ Failures/10^6 Hours

G) Thick Film Resistors, per instructions in Section 5.5, the contribution of these devices is considered insignificant relative to the overall hybrid failure rate and they may be ignored.

Overall Hybrid Part Failure Rate Calculation:

$$\lambda_p = [\Sigma N_C \lambda_c] (1 + .2 \pi_E) \pi_F \pi_Q \pi_L$$

π_E	= 6.0	Section 5.10
π_F	= 5.8	Section 5.5
π_Q	= 1	Section 5.10
π_L	= 1	Section 5.10

$\lambda_p = [(1)(.038) + (1)(.031) + (2)(.0023) + (2)(.0023)$
$+ (2)(.0069) + (2)(.0039)](1 + .2(6.0))(5.8)(1)(1)$

$\lambda_p = 1.3$ Failures/10^6 Hours